Journal of Building Engineering 103 (2025) 112126



Contents lists available at ScienceDirect

Journal of Building Engineering

journal homepage: www.elsevier.com/locate/jobe





Digital twin safety analysis of large-span spatial structure based on airborne LiDAR

Kaichao Wang [a], Chunxia Zhang [b], Yachao Qian [c], Wulang Huang [d], Tianqi Yi [c], Jian Wang [d], Desheng Huang [e], Xiaoxiong Zha [a,*]

[a] School of Civil and Environmental Engineering, Harbin Institute of Technology, Shenzhen, Shenzhen, 518055, China
[b] Design Management Center, Bureau of Public Works of Shenzhen Municipality Engineering, Shenzhen, 518027, China
[c] Yuanyu Smart Data (Shenzhen) Technology Co.,Ltd., Shenzhen, 518063, China
[d] China Construction Science and Industry Corporation Ltd., Shenzhen, 518054, China
[e] China Construction Steel Structure Engineering Co., Ltd., Shenzhen, 518118, China

ARTICLE INFO

Keywords:
Airborne LiDAR
Spatial structure
Geometric digital twin
Initial geometric defects
Safety analysis

ABSTRACT

This paper introduces a novel method for assessing the safety of post-construction large-span spatial structures. Initially, a brief overview is provided of the operating principles and measurement equipment associated with airborne Light Detection and Ranging (LiDAR) technology. Subsequently, a methodology for extracting one-dimensional geometric models from point cloud data is described. The proposed safety analysis method is then applied to the retractable roof of the Shenzhen Gymnasium, with a comparison made between two approaches: the LiDAR-based method and the consistent imperfection mode method, in terms of their consideration of the impact of initial geometric imperfections on structural mechanical properties. The research findings reveal that the one-dimensional geometric models extracted from airborne LiDAR measurements can accurately reflect the real-world structural imperfections and can be regarded as geometric digital twins of the structures. In comparison with the consistent imperfection mode method, finite element models based on airborne LiDAR measurement technology offer a superior representation of the real performance of the structures.

1. Introduction

The use of large-span spatial structures in large-scale public buildings is increasing, due to the advantages of light weight, excellent mechanical performance, a variety of structural forms, innovative shapes and substantial available space. Examples of such structures include the Shenzhen Gymnasium, Shenzhen Financial and Cultural Centre, Shenzhen Natural History Museum and Shenzhen Reform and Opening Exhibition Hall, which are currently under construction and are about to start construction. However, the large-span spatial structure mostly adopts steel structure, which is susceptible to instability and failure of components or structures. The initial defects of the components during the production and transportation processes, the asymmetrical docking of components, and the deviation of joints caused by the on-site installation process can exacerbate the instability and failure of the steel structure. Consequently, the impact of initial geometric imperfections must be considered during the stability analysis of large-span spatial structures.

* Corresponding author. School of Civil and Environmental Engineering, Harbin Institute of Technology, Shenzhen, 518055, China.
E-mail address: zhahero@126.com (X. Zha).

https://doi.org/10.1016/j.jobe.2025.112126
Received 13 July 2024; Received in revised form 27 January 2025; Accepted 12 February 2025
Available online 13 February 2025
2352-7102/© 2025 Published by Elsevier Ltd.

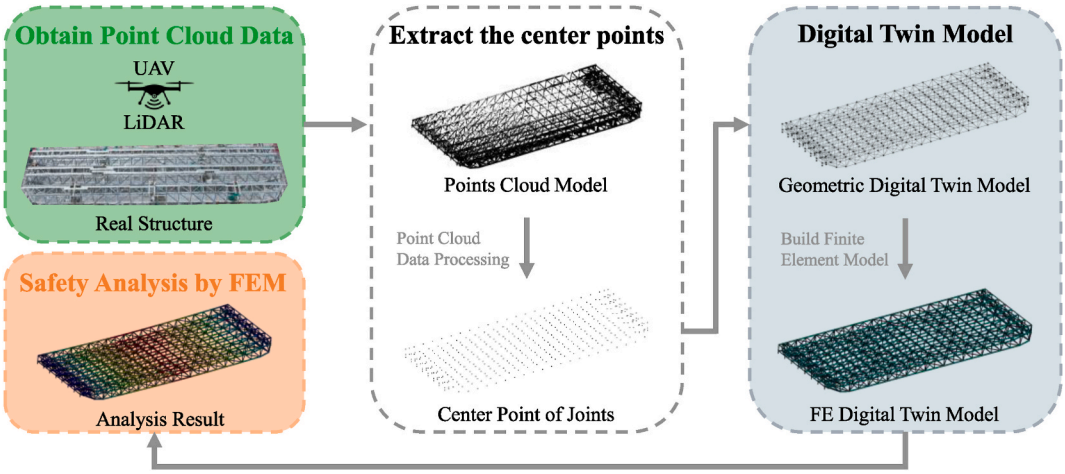


Fig. 1. Framework for digital twin safety analysis of large-span spatial structure based on airborne LiDAR.

Regarding the influence of initial geometric imperfections on the stability performance of large-span spatial structures, Ma et al. [1] investigated the effect of different defect shapes and amplitudes on the stability bearing capacity of a stadium roof structure by considering the first three buckling modes and their combinations as the defect distribution. Du [2] investigated the effect of different defect modes on the overall stability of a roof structure by considering the first six buckling modes as the defect distribution. Wang [3] investigated the overall stability performance of a portal frame structure system by taking the first elastic buckling mode as the defect distribution. Jiang et al. [4] investigated the influence of different defect sizes on the overall stability performance of a retractable roof structure using the first buckling mode as the defect distribution. Díaz et al. [5] investigated the overall stability performance of large-span metal roofs by considering the first mode of buckling analysis as the defect distribution. Liu et al. [6] proposed a random defect mode superposition method for the stability analysis of single layer reticulated shell structures. This method identifies the most unfavourable defect mode for structural instability. Xu et al. [7] introduced a reliability-based method for calculating the critical stability load-bearing capacity of single-layer reticulated shell structures and validated its effectiveness through two case studies. Borri et al. [8] used the random defect mode method to study the influence of the distribution pattern and size of initial geometric imperfections on the structural stability bearing capacity. Chen et al. [9] proposed a modelling method for initial geometric imperfections in reticulated shell structures based on random field theory and used it to investigate the effect of these imperfections on the structural stability performance. Furthermore, Cui et al. [10] proposed a simultaneous autoregressive model capable of simulating spatially varying initial geometric imperfections, including those in irregular reticulated shell structures. As can be seen from the above studies, the main methods currently used to account for initial geometric imperfections are the random defect mode method and the consistent defect mode method. However, it should be noted that the defect modes assumed in these methods are random, whereas in constructed structures the defect mode is deterministic. Therefore, when performing a safety analysis on an existing structure, calculations should be based on the actual defect shape.

Digital twins, which serve as digital replicas of real-world assets, can effectively reflect the actual state of structures, including stresses, deformations and geometric imperfections. Many scholars have used three-dimensional (3D) laser scanning technology and reverse modelling techniques to capture geometric imperfections of structures and create geometric digital twins. Sangani et al. [11] used 3D digital image correlation equipment to scan tubular steel columns with initial geometric defects, and evaluated the residual capacity of tubular steel columns according to the scanned model. Chacon et al. [12] measured the initial geometric defects of planar steel frames using a terrestrial laser scanner. Selvaraj [13] and Zhang [14] demonstrated that 3D laser scanning can be used to measure the initial geometric defects of steel members. Xu et al. [15] used 3D laser scanning technology to detect the full-field geometric defects of circular steel pipes and investigated the influence of geometric defects on the compressive and flexural strength of circular steel pipes. Tzortzinis et al. [16] used 3D laser scanning technology to measure the corrosion of bridges and combined 3D laser scanning with finite element to evaluate the residual capacity of steel beams. Ban et al. [17] used 3D laser scanning technology to measure the size and geometric defects of square hollow section short columns, and investigated the local buckling behaviour of square hollow section short columns. Lazzari et al. [18] proposed a method for measuring geometric defects and dimensional deviations in GFRP profiles using a 3D contact coordinate measuring machine. Feng et al. [19] used 3D laser scanning technology to obtain and evaluate initial geometric defects such as prestressed CFRP reinforced columns. Zou et al. [20] attempted to use 3D laser scanning technology to obtain the initial geometric defects of steel structural components, and proposed a method to build a digital model of the components based on 3D laser point cloud data. Xu et al. [21] proposed a method to determine the initial geometric defects of steel members and establish a geometric model by using 3D scanning, and proved that this method can better predict the initial geometric defects of steel members. Aktepe et al. [22,23] used a 3D laser scanner to measure the geometric defects of cold-formed thin-walled steel in detail to propose a predictive expression for the initial geometric defects. The experimental results show that the finite element results based on the scanning model are in good agreement with the experimental results. Ma et al. [24] described a method that combines two types of laser scanning technologies (terrestrial laser scanning and unmanned aerial vehicle (UAV)-based laser scanning) for detecting the

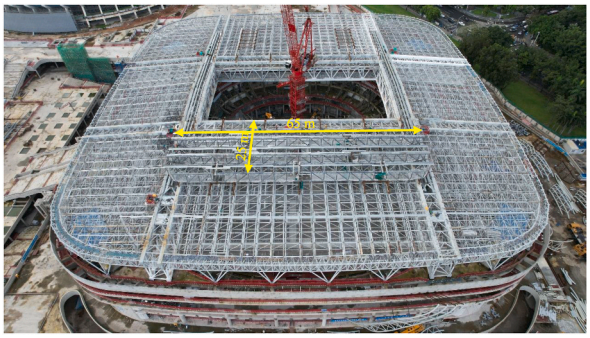


Fig. 2. Schematic diagram of gymnasium.

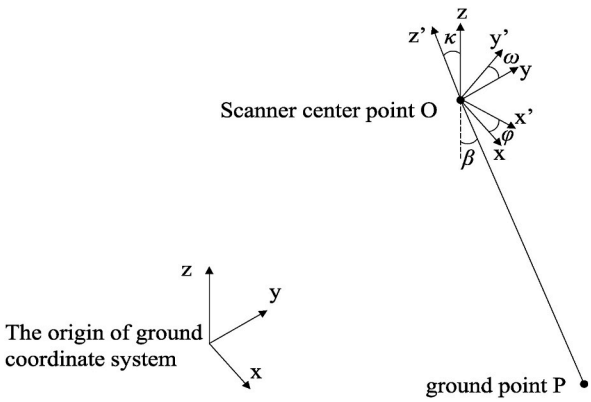


Fig. 3. Positioning principle diagram of airborne lidar measurement system [29].

dimensional quality of bridges after construction. Lu et al. [25] proposed an automated approach to create geometric digital twin models of bridges based on point cloud data. Drobnyi et al. [26] presented a method for automatically detecting connectivity relationships in point clouds of buildings, with the aim of improving the efficiency of constructing geometric digital twin models. Both Rúa et al. [27] and Hermida [28] proposed methods for extracting variable geometric lines from point cloud data, specifically extracting geometric lines from corroded steel beams and wooden truss structures, respectively. These extracted lines can be used for structural calculations and analysis. The above studies demonstrate that 3D laser scanning technology can accurately capture the true initial geometric imperfections of components and create geometric digital twins of structures. However, most current research uses terrestrial 3D laser scanners to acquire point cloud data of structures, and the geometric digital twins produced are mostly in the form of 3D solid models. Terrestrial 3D laser scanners face challenges in measuring spatial structures with large spans, and it is not economically feasible to use 3D solid models for computational analysis of such structures.

Therefore, in order to accurately obtain the real geometric defects of the large-span spatial structure and reduce the computational effort of the finite element model, this paper proposes a method to obtain the real geometric defects of the structure by using the airborne LiDAR measurement technology. At the same time, the method of extracting a one-dimensional geometric model from point cloud data is described. Taking the movable roof of Shenzhen Gymnasium as an example, a finite element model based on LiDAR is established, and the difference of finite element results under different initial geometric defect modes is explored. The framework of digital twin safety analysis of large-span spatial structure based on airborne LiDAR is shown in Fig. 1.

2. Engineering overview

The Shenzhen Gymnasium has a total of 6 floors, with a total floor area of 57,600 square metres and a building height of 45.00 m. It consists of a steel roof structure and a lower grandstand. The gymnasium is shown in Fig. 2. The lower grandstand adopts the reinforced concrete frame-shear wall structural system, and the lower grandstand is the supporting structure of the roof steel structure canopy. The roof steel structure canopy is a radial retractable roof composed of one fixed roof and two movable roofs. Each movable roof is supported by eight trolleys above the fixed roof. The fixed roof and the movable roof are mainly composed of orthogonal plane truss system, where the span of the movable roof is 65 m and the width is 25 m.

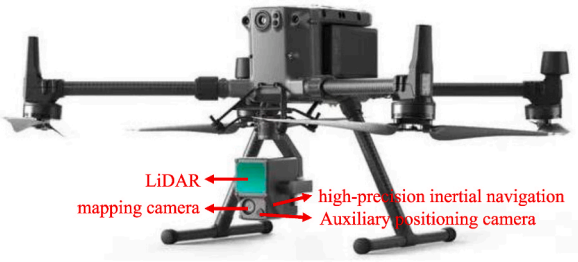


Fig. 4. Picture of Zenmuse L1.

Table 1
Main technical parameters of Zenmuse L1.

Name	Item	Parameter
System performance	Range	450m @ 80 %, 0 klx 190m @ 10 %, 100 klx
	Point cloud data rate	Single echo: maximum 240000 points/second Multi-echo: maximum 480000 points/second
	Systematic precision	horizontal accuracy: 10 cm @ 50m elevation accuracy: 5 cm @ 50m
LiDAR	Accuracy of ranging	3 cm @ 100m
	Support the maximum number of echoes	3
	FOV	repeat scanning:70.4° × 4.5° Non-repetitive scanning:70.4° × 77.2°
Inertial navigation system	IMU refresh rate	200 Hz
	Heading accuracy	real-time:0.3° Post processing:0.15°
	Pitch/roll accuracy	real-time:0.05° Post processing: 0.025°
Auxiliary positioning camera	Resolution	1280 × 960
	FOV	95°
Mapping camera	Sensor size	1 inch
	Active picture elements	20 million
	Dimension of image	5472 × 3078 (16: 9) 4864 × 3648 (4: 3) 5472 × 3648 (3: 2)
	Focal length	8.8 mm/24 mm (equivalence)

3. Airborne LiDAR measurement principle and equipment introduction

3.1. Measurement principle

Airborne LiDAR measurement system is an active measurement system composed of aircraft, dynamic differential global positioning system, inertial navigation system and laser rangefinder. It takes the aircraft as the carrying platform of the whole measurement system, uses the dynamic differential global positioning system to record the coordinate position of the aircraft, uses the inertial navigation system to record the navigation parameters such as the attitude, speed and position of the aircraft, and uses the laser ranging system to measure the distance and orientation between the aircraft and the measurement object.

The principle of determining the coordinates of the target by the airborne LiDAR system is shown in Fig. 3. The points O and P on the map are the position of the aircraft and the position of the target, respectively. The position coordinates O (x_0, y_0, z_0) of the aircraft are recorded in real time by the dynamic differential global positioning system. The pitch angle σ, roll angle ω and yaw angle k of the aircraft are recorded in real time by the inertial navigation system. The distance R and azimuth β between the aircraft and the measurement point are recorded in real time by the laser rangefinder. Therefore, the position coordinates P (x_p, y_p, z_p) of the measurement point can be calculated according to equation (1) and (2) [29].

$$\begin{cases} x_p = x_0 + \Delta x_p \\ y_p = y_0 + \Delta y_p \\ z_p = z_0 + \Delta z_p \end{cases} \tag{1}$$

$$\begin{cases} \Delta x_p = f_x(\varphi, \omega, \kappa, \beta, R) \\ \Delta y_p = f_y(\varphi, \omega, \kappa, \beta, R) \\ \Delta z_p = f_z(\varphi, \omega, \kappa, \beta, R) \end{cases} \tag{2}$$

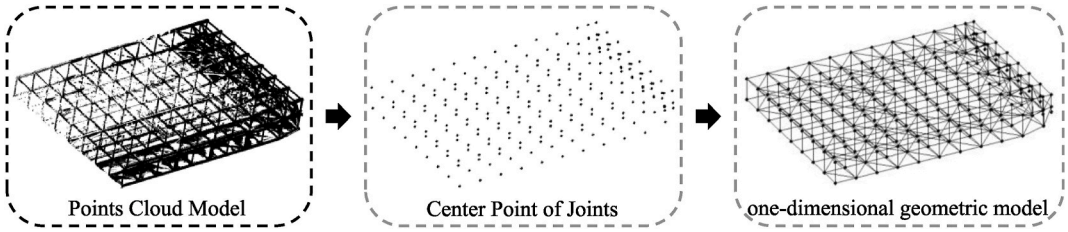


Fig. 5. Extraction process of one-dimensional geometric model.

3.2. Equipment

This project uses Zenmuse L1 for scanning, as shown in Fig. 4. Zenmuse L1 is an integrated industrial laser load that integrates LIVOX™ LiDAR, high-precision inertial navigation and mapping camera, and can quickly generate high-precision point clouds and models. The Zenmuse L1 is equipped with a three-axis pan-tilt camera that can be installed on the DJ designated flight platform and supports the DJI Pilot App. Together with DJI Terra software, it provides an integrated solution for 3D scene reconstruction and high-precision 3D reconstruction of complex scenes in all weather conditions, efficiently and in real time. The main technical parameters of the Zenmuse L1 are listed in Table 1.

The main tasks of using Zenmuse L1 to collect point cloud data outdoors are as follows: 1. Make sure that the L1 load is correctly installed on the aircraft, to avoid the L1 load falling in the operation project; 2. Open the power supply of the aircraft and the remote control one by one, and make sure that the aircraft and the remote control complete the frequency matching; 3. Enter the DJI Pilot App manual flight interface, select the corresponding RTK service type, and make sure that the positioning and orientation state of RTK is FIX; 4. The L1 load is preheated for 3–5 min after starting, to make sure that the high-precision inertial navigation is in a stable working temperature environment; 5. According to the light conditions in the field, set the camera parameters, select the camera focusing model, Zenmuse L1 supports MF (manual focusing), AFC (automatic continuous focusing) and AFS (automatic single point focusing); 6. Select the route task type, Zenmuse L1 supports four types of flight including flight, aerial photography, ground flight and manual flight, and set the task parameters; 7. Save the mission, upload the route and execute the mission. During mission execution, the point cloud can be paused or started to record at any time, and the point cloud results can be previewed in real time during the scanning process; 8. Turn off the aircraft power after the mission.

It should be noted that: 1. During the data collection process, it is imperative that the aircraft is in motion and the flight speed exceeds 2 m per second. In the absence of these conditions, DJI Terra will reconstruct errors in data post-processing. 2. Following 100 s of uninterrupted flight, it is necessary to manually accelerate or decelerate, or alternatively click ‘Calibration flight’ to calibrate the inertial navigation. 3. Following the conclusion of data acquisition, manual inertial navigation calibration is mandatory. L1 will automatically record the inertial navigation data for approximately 2 min after the point cloud acquisition for DJI Terra’s posterior difference decomposition calculation.

4. Point cloud data processing

In the initial phase, obvious noise points are manually removed. Subsequently, Gaussian filtering and the K-means algorithm are applied, respectively, to eliminate noise points between structural members. It is crucial to preserve the geometric feature information of the roof structure during the noise reduction process.

4.1. Extraction method of one-dimensional geometric model

The process of extracting the one-dimensional geometric model from the point cloud is shown in Fig. 5. First, the centre point coordinates of the joints in the roof are extracted according to the point cloud model, and then these centre points are connected by straight lines according to the geometric relationship of the structural members.

Before extracting the centroid of the joints, it is necessary to identify the optimal valid cylinder representation for each component based on the point cloud data. As we know, a cylinder in three-dimensional space can be defined by a point c (c_x, c_y, c_z) on its central axis, the unit normal vector $\vec{v} = (v_x, v_y, v_z)$ of the central axis and the radius r of the cylinder. Consequently, n points are randomly selected from the point cloud model and fitted, and an optimal cylinder equation for these n points is obtained through iterative calculations. Subsequently, the optimal cylinder is substituted into the point cloud dataset to determine its validity: if many points lie on this cylinder, the cylinder is considered valid; if only a few points lie on the cylinder, it is considered invalid. After obtaining valid cylinders, the points falling on these valid cylinders are removed from the point cloud dataset. The above process is then repeated to fit the remaining points until all valid cylinders are identified. The optimal cylinder equation for n points is obtained by iteratively solving for the minimum value of equation (3), where $\overrightarrow{p_i}$ denotes the vector from the origin to a point P in the point cloud dataset, $\overrightarrow{c}$ denotes the vector from the origin to a point c on the central axis of the cylinder, and the constant λ is a small value.

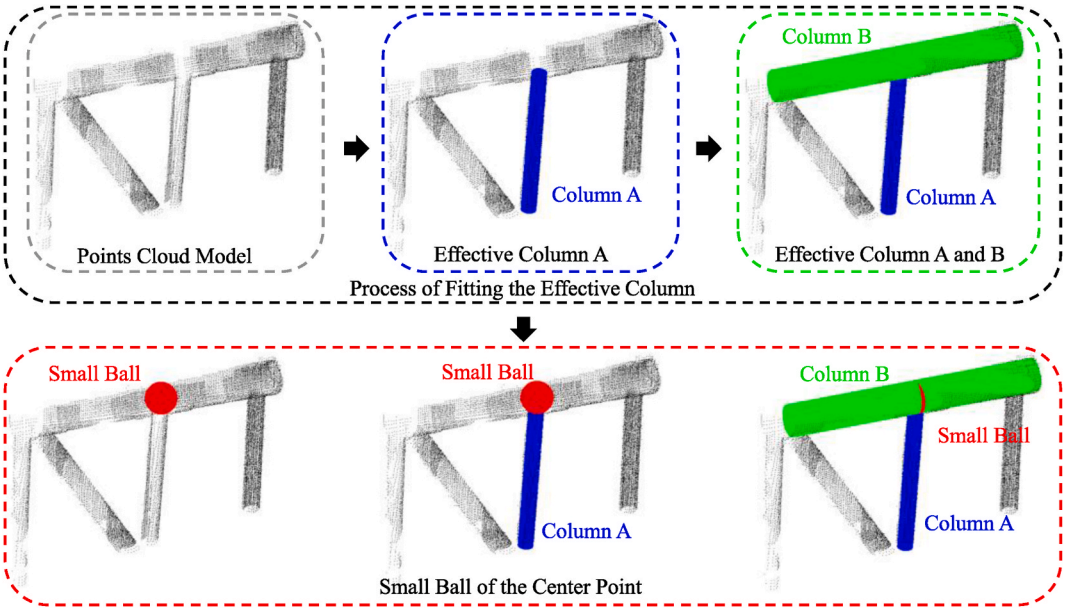


Fig. 6. Schematic diagram of fitting effective cylinders and joint ball.



Fig. 7. Scan photos of the scene.

$$\mathscr{L} = \sum_i \left[(\vec{p_i} - \vec{c} - [(\vec{p_i} - \vec{c}) \cdot \vec{v}]\vec{v})^2 - r^2 \right]^2 - \lambda\left(|\vec{v}|^2 - 1\right) \tag{3}$$

The centre points of the joints in the roof structure are extracted based on valid cylinders. From the valid cylinders, the vector for any point on the central axis of the cylinder can be obtained as shown in equation (4), where ω is an arbitrary constant. By evaluating equation (5), the two nearest points C_A and C_B on the central axis of the valid cylinder A and valid cylinder B respectively can be determined. In this study, the centre of C_A and C_B is taken as the centre of the joint at the intersection of valid cylinder A and valid cylinder B.

$$\vec{C} = \vec{c} + \varpi\vec{v} \tag{4}$$

$$\min \left(\overrightarrow{C_A} - \overrightarrow{C_B}\right)^2 \tag{5}$$

Fig. 6 Presents a flowchart illustrating the process of fitting valid cylinders from point cloud data, along with the depiction of small sphere based on the central point of joint. As evident from the figure, Cylinder A and Cylinder B fit well with the point cloud data, with the small sphere located at the intersection points of valid Cylinder A and valid Cylinder B. This demonstrates that the method proposed in this section can effectively extract the central points of joints in the roof structure from the point cloud dataset.



Fig. 8. Point cloud scanned by Zenmuse L1.

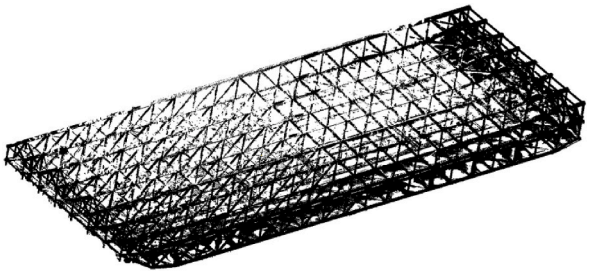


Fig. 9. Point cloud of single movable roof.

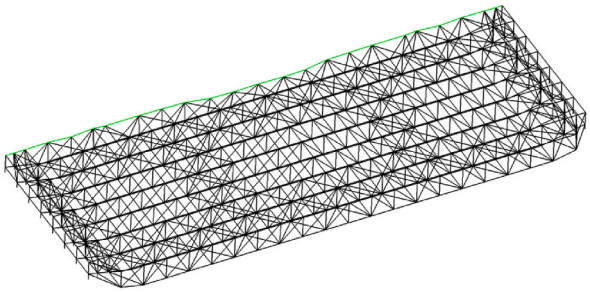


Fig. 10. One-dimensional geometric model of single movable roof.

The one-dimensional geometric model obtained by this method has been shown to offer a more accurate reflection of the structure's true geometric shape and initial geometric defects. Consequently, the one-dimensional geometric model extracted from the airborne LiDAR measurement results can be regarded as the digital twin of the structure in geometry.

4.2. Results of airborne LiDAR scanning

In this project, when collecting point cloud data outdoors, the camera focus model is selected as AFC, the route task type is selected as mapping aerial photography, the camera is selected as LiDAR Mapping, the calibration flight option is turned on, the flight height is set to 80 m, the flight speed is set to 10 m/s, the laser side overlap rate is set to 50 %, the main route angle is set to 150°, the side distance is set to 25 m, the double echo mode is selected, the sampling frequency is set to 240 KHz, the scanning mode is selected as non-repetitive scanning, the true color is turned on, and other parameters remain default.

Subsequent to the completion of the scanning process, the DJI Terra software, which is compatible with DJI L1, is utilised for the purpose of processing the scanning results. The scanned point cloud is displayed in Fig. 7. The point cloud data of a single movable roof is then extracted from Fig. 8, as illustrated in Fig. 9.

Fig. 10 presents the one-dimensional geometric model of the movable roof, wherein the coordinates of nodes along the green linear entities are determined from point cloud data, while the coordinates of other nodes are established based on design values. Due to limitations in measurement conditions, time, and equipment, distortions are observed in the point cloud data within the truss structure. Consequently, the methodology outlined in Section 4.1 is solely applied to extract the coordinates of nodes located on the green linear entities.

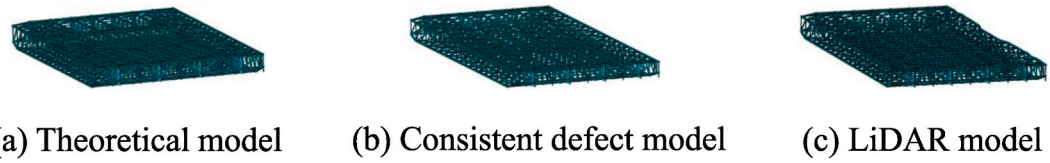


Fig. 11. Three different analysis models.

Table 2
Material parameters of Q355B.

Thickness/mm	Elastic modulus/MPa	Yield strength/MPa	Tangent modulus/MPa	Ultimate tensile strength//MPa
t < 16	206000	345	2060	470
16<t < 40	206000	335	2060	470

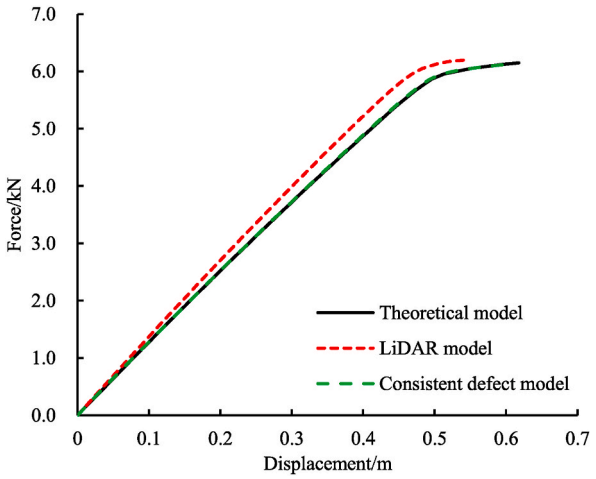


Fig. 12. Relationship curve between mid-span displacement and nodal load.

5. Safety analysis by FEM

To explore the differences between point cloud model and theoretical model, and to compare the differences between the two methods (airborne LiDAR method and consistent defect mode method) in considering the impact of initial geometric defects on structural performance, this paper established theoretical model, LiDAR model, and consistent defect model, respectively. The three finite element models are shown in Fig. 11. Among them, the theoretical model is the model used in structural design; the LiDAR model is a model that modifies the coordinate values of some nodes according to the LiDAR scanning results; the consistent defect model refers to a model where the defect size is 1/300 of the span of the movable roof, and the defect form is the first buckling mode under gravity.

5.1. Finite element modelling method

When the finite element model is established, the movable roof adopts BEAM189 element with a length of 1 m. All the members are connected by common nodes, and the translational displacement of the node XYZ in three directions at the trolley position is constrained. Q355B steel is used in the movable roof, and the steel in the finite element is bilinear elastic-plastic constitutive. The material parameters are shown in Table 2. During the analysis, the node load is applied on the nodes on the top of the roof, and the bearing capacity of the movable roof is determined by gradually increasing the size of the node load.

5.2. Results

The relationship curve between the mid-span displacement and the node load of the three models is shown in Fig. 12. The yield points of the theoretical model, LiDAR model, and consistent defect model determined by the farthest point method [30] are (0.552, 6.052), (0.503, 6.130) and (0.544, 6.065), respectively. The initial slopes of the three models are 12.528 kN/m, 13.778 kN/m and 12.844 kN/m, respectively. As shown in the figure, the curves of the theoretical model and the consistent defect model basically

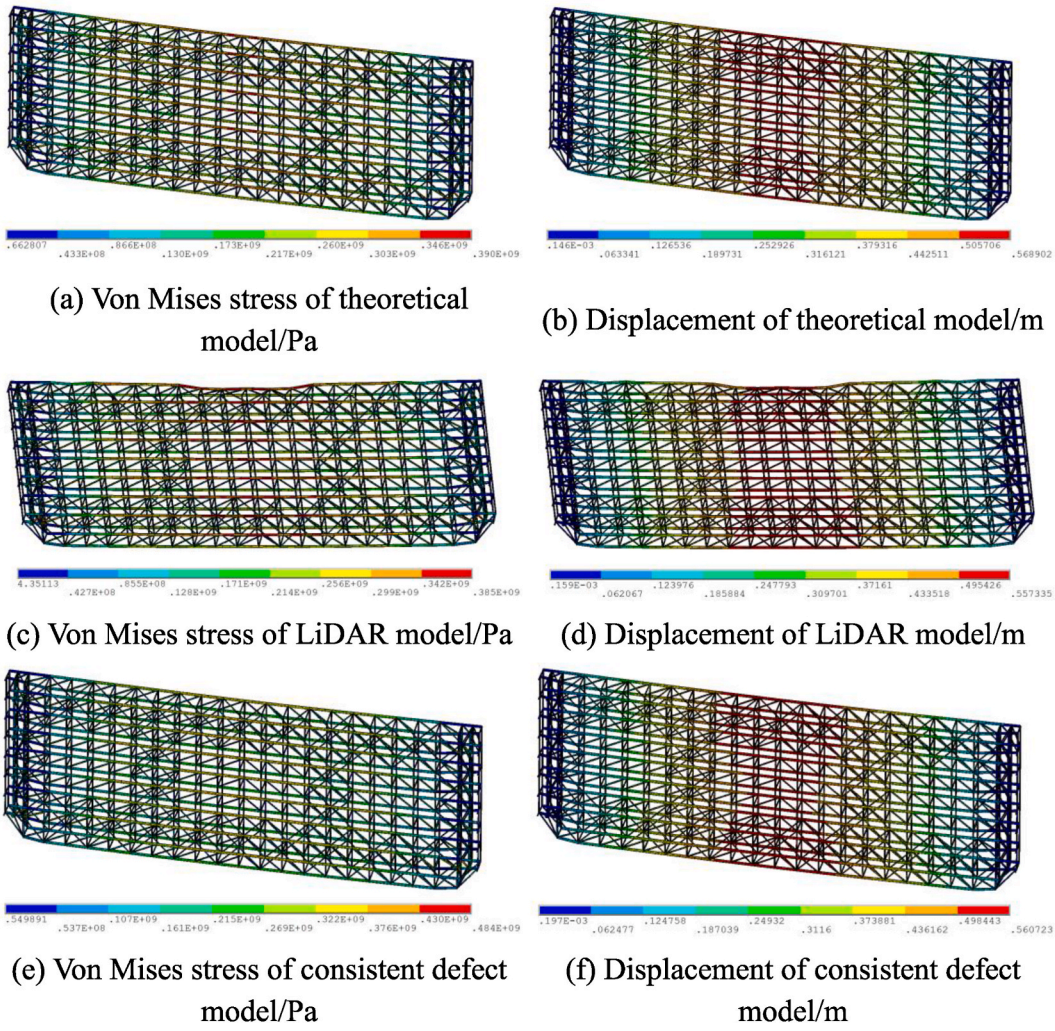


Fig. 13. Finite element results.

overlap, while the curves of the LiDAR model have certain differences from the theoretical model and the consistent defect model. The yield loads of the LiDAR model and the consistent defect model are 1.013 times and 1.002 times that of the theoretical model, respectively. The initial deformation stiffness of the LiDAR model and the consistent defect model are 1.100 times and 1.025 times that of the theoretical model, respectively. The ultimate bearing capacity of the LiDAR model and the consistent defect model are 1.007 times and 0.996 times that of the theoretical model, respectively.

When the load reaches the yield point, the von Mises equivalent stress and deformation of the three models are shown in Fig. 13. It can be seen from the figure that the maximum displacement of the three models is located on the upper chord of the roof span, and the maximum equivalent stress is near the trolley. When the node load reaches the yield point, the maximum equivalent stress of the three models has exceeded the yield strength of Q355B. The maximum equivalent stress of the LiDAR model and the consistent defect model are 0.987 times and 0.985 times of the theoretical model, respectively. The maximum displacement of the LiDAR model and the consistent defect model are 0.980 times and 0.986 times of the theoretical model, respectively.

Through the above comparison, it can be found that there are certain differences in stress and deformation between the actual structure after completion and the theoretical design model. The finite element model based on LiDAR can better reflect the influence of initial geometric defects on the mechanical properties of the structure than the finite element model based on the consistent defect mode method.

6. Conclusion

In order to evaluate the safety of large-span space structures after completion and explore the impact of actual initial geometric defects on the mechanical properties of large-span space structures, this paper combines airborne LiDAR measurement technology with structural safety analysis. The main research results are as follows.

1. A method of extracting one-dimensional geometric model from point cloud model is proposed. The geometric model based on this method can reflect the real geometric shape and initial defect of the structure. It can be regarded as the digital twin of the actual structure in geometry.
2. Propose a digital twin safety analysis method that combines airborne LiDAR measurement technology with structural safety analysis. Compared to safety analysis based on theoretical models, the results obtained using this method are more accurate; Compared to the uniform buckling mode method, the finite element model based on LiDAR can better reflect the influence of initial geometric defects on the mechanical properties of the structure.

CRediT authorship contribution statement

Kaichao Wang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Chunxia Zhang:** Supervision, Resources, Investigation, Data curation. **Yachao Qian:** Writing – original draft, Visualization, Validation, Software, Data curation. **Wulang Huang:** Validation, Software, Investigation, Data curation. **Tianqi Yi:** Writing – original draft, Validation, Software, Data curation. **Jian Wang:** Validation, Supervision, Resources, Investigation. **Desheng Huang:** Supervision, Resources, Investigation, Formal analysis. **Xiaoxiong Zha:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] Z. Ma, X. Mao, M. Li, C. Liu, Research on nonlinear stability of long-span pipe truss roof considering initial geometric imperfections, Struct. Eng. 38 (1) (2022) 47–52, https://doi.org/10.15935/j.cnki.jggcs.2022.01.002.
[2] Z. Du, Stability analysis of ruijin sports center stadium steel roof, Build. Struct. 47 (S1) (2017) 707–711, https://doi.org/10.19701/j.jzjg.2017.s1.155.
[3] Z. Wang, Global stability analysis of the central steel structure of wanda-mall project, Steel Construct. 30 (5) (2015) 36–39+78, https://doi.org/10.13206/j.gjg201505008.
[4] Jiang X, Chen Y, Liu K, Chen Y. Global stability analysis of retractable steel roof in Nantong stadium. Build. Struct. 2008(9); 20-23+46. DOI:10.19701/j.jzjg.2008.09.005.
[5] J. Díaz, P. Nieto, J. Vilán, J. Sierra, Non-linear buckling analysis of a self-weighted metallic roof by FEM, Math. Comput. Model. 51 (2010) 216–228, https://doi.org/10.1016/j.mcm.2009.08.032.
[6] H. Liu, Y. Luo, L. Yang, W. Zhang, Stochastic imperfection mode superposition method for stability analysis of single-layer lattice domes, J. Tongji Univ. 40 (9) (2012) 1294–1299.
[7] J. Xu, H. Wang, Y. He, hua Dong, Reliability-based method for determining of critical stable bearing capacity of single-layer dome structures, J. Build. Struct. 45 (1) (2024) 76–84, https://doi.org/10.14006/j.jzjgxb.2022.0342.
[8] C. Borri, P. Spinelli, Buckling and post-buckling behaviour of single layer reticulated shells affected by random imperfections, Comput. Struct. 30 (4) (1988) 937–943, https://doi.org/10.1016/0045-7949(88)90131-9.
[9] G. Chen, H. Zhang, J. Kim, F. Fan, Modeling geometric imperfections for reticulated shell structures using random field theory, Eng. Struct. 126 (2016) 481–489, https://doi.org/10.1016/j.engstruct.2016.08.008.
[10] X. Cui, Y. Li, H. Hong, Effect of spatially correlated initial geometric imperfection on reliability of spherical latticed shell considering global instability, Struct. Saf. 82 (2020) 101895, https://doi.org/10.1016/j.strusafe.2019.101895.
[11] S. Prithvi, A. Anil, A comprehensive methodology for residual capacity evaluation of damaged steel tubular columns, J. Constr. Steel Res. 216 (2024) 108625, https://doi.org/10.1016/j.jcsr.2024.108625.
[12] C. Rolando, P. Carol, R. Esther, TLS measurements of initial imperfections of steel frames for structural analysis within BIM-enabled platforms, Autom. ConStruct. 125 (2021) 103618, https://doi.org/10.1016/j.autcon.2021.103618.
[13] S. Selvaraj, M. Madhavan, Geometric imperfection measurements and validations on cold-formed steel channels using 3D noncontact laser scanner, J. Struct. Eng. 144 (3) (2018) 04018010, https://doi.org/10.1061/(ASCE)ST.1943-541X.0001993.
[14] Z. Zhang, G. Shi, L. Hou, et al., Geometric dimension and imperfection measurements of box-T section columns using 3D scanning, J. Constr. Steel Res. 183 (2021) 106742, https://doi.org/10.1016/j.jcsr.2021.106742.
[15] Y. Xu, B. Wu, B. Zheng, Full-field geometric imperfection and effect on cross-section capacity of circular steel tubes, J. Constr. Steel Res. 201 (2023) 107749, https://doi.org/10.1016/j.jcsr.2022.107749.
[16] G. Tzortzinis, C. Ai, S. Breña, S. Gerasimidis, Using 3D laser scanning for estimating the capacity of corroded steel bridge girders: experiments, computations and analytical solutions, Eng. Struct. 265 (2022) 114407, https://doi.org/10.1016/j.engstruct.2022.114407.
[17] H. Ban, Y. Mei, Local buckling behaviour of stainless-clad bimetallic steel built-up square hollow section stub columns, Thin-Walled Struct. 182 (2023) 110207, https://doi.org/10.1016/j.tws.2022.110207.
[18] J. Lazzari, L. Martins, N. Silvestre, Global and local geometrical imperfections of pultruded GFRP profiles based on a modal approach, Constr. Build. Mater. 412 (2024) 134813, https://doi.org/10.1016/j.conbuildmat.2023.134813.
[19] P. Feng, Y. Zou, L. Hu, T. Liu, Use of 3D laser scanning on evaluating reduction of initial geometric imperfection of steel column with pre-stressed CFRP, Eng. Struct. 198 (2019) 109527, https://doi.org/10.1016/j.engstruct.2019.109527.
[20] Y. Zou, L. Hu, P. Feng, Q. Li, M. Teng, Precise measurement method of initial geometric imperfections of steel structural members using three-dimensional laser scanning, Ind. Constr. 51 (5) (2021) 22–29, https://doi.org/10.13204/j.gyjzG20110602.
[21] D. Xu, Y. Wang, X. Liu, V. Chen, Y. Bu, A novel method and modelling technique for determining the initial geometric imperfection of steel members using 3D scanning, Structures 49 (2023) 855–874, https://doi.org/10.1016/j.istruc.2023.01.136.
[22] R. Aktepe, B. Erkal, Prediction of the initial geometric imperfection magnitudes for numerical modeling of cold-formed steel channel sections, Structures 60 (2024) 105869, https://doi.org/10.1016/j.istruc.2024.105869.

[23] R. Aktepe, B. Erkal, Experimental and numerical study on flexural behaviour of cold-formed steel hat-shaped beams with geometrical imperfections, J. Constr. Steel Res. 202 (2023) 107774, https://doi.org/10.1016/j.jcsr.2023.107774.
[24] M. Cabaleiro, B. Riveiro, P. Arias, J. Caamaño, J. Vilán, Automatic 3D modelling of metal frame connections from LiDAR data for structural engineering purposes, ISPRS J. Photogrammetry Remote Sens. 96 (2014) 47–56, https://doi.org/10.1016/j.isprsjprs.2014.07.006.
[25] R. Lu, I. Brilakis, Digital twinning of existing reinforced concrete bridges from labelled point clusters. Proceedings of the 36th International Symposium on Automation and Robotics in Construction (ISARC), 2019, https://doi.org/10.22260/ISARC2019/0082.
[26] V. Drobnyi, S. Li, I. Brilakis, Connectivity detection for automatic construction of building geometric digital twins, Autom. ConStruct. 159 (2024) 105281, https://doi.org/10.1016/j.autcon.2024.105281.
[27] E. Rúa, M. Cabaleiro, B. Conde, B. Riveiro, First results of a methodology to obtain a 1D variable geometry model for the structural analysis of corroded steel beams from the point cloud, Structures 33 (2021) 3257–3268, https://doi.org/10.1016/j.istruc.2021.06.063.
[28] J. Hermida, M. Cabaleiro, B. Riveiro, J. Caamaño, Two-dimensional models of variable inertia from LiDAR data for structural analysis of timber trusses, Constr. Build. Mater. 231 (2020) 117072, https://doi.org/10.1016/j.conbuildmat.2019.117072.
[29] S. Lang, Multi-temporal Airborne LiDAR Data-Based Object-Level 3D Building Changes Detection, Liaoning Technical University, 2023, https://doi.org/10.27210/d.cnkiglnju2023.000631.
[30] P. Feng, H. Qiang, L. Ye, Discussion and definition on yield points of materials, members and structures, Eng. Mech. 34 (2017) 36–46, https://doi.org/10.6052/j.issn.1000-4750.2016.03.0192.